

Notes: Gu, Peng, and Zhang (RFS, 2024)

The Gender Gap in Household Bargaining Power: A Revealed-Preference Approach

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1 Big picture

- **Question.** When husbands and wives have different risk preferences, whose preference matters more for household portfolio choice, and what determines that intrahousehold bargaining power?
- **Setting.** The paper develops a structural model of household portfolio choice with bargaining over risk preferences and estimates it primarily on Australian panel data from the HILDA Survey, then checks the main patterns in Germany and the United States.
- **Core challenge.** Standard household-finance work usually treats the household as a single decision-maker or uses a survey proxy such as the “financial head” or “final say” question. That makes it hard to connect individual preferences to actual household choices and hard to quantify the relative importance of different determinants of bargaining power.
- **Main methodological contribution.** The paper proposes a **revealed-preference** approach: if spouses’ individual risk preferences are observable, then the household’s stock-market participation and risky-asset holdings reveal how much weight the household places on each spouse’s preference. The model then maps those weights into bargaining power.
- **Why this matters.** The paper links three things that are often studied separately: individual risk preferences, household portfolio choice, and gendered bargaining power. It also provides a structural alternative to reduced-form and survey-based approaches.
- **Main empirical question.** How large is the gender gap in bargaining power inside the household, how much of it is explained by observed spousal differences such as earnings and employment, and how much is due to gender effects and gender norms?
- **Main empirical results.**
 - In Australia, the average household places about **60%** weight on the husband’s risk preference and **40%** on the wife’s.
 - Roughly **half** of this Australian gap is explained by observable spousal differences, especially **earnings** and **employment**; the other half is a residual **gender effect**.
 - Gender effects are strongest in **husband-headed** households and much smaller in wife-headed households.
 - Households with more **traditional gender norms** place more bargaining power on husbands and are more likely to be husband-headed in financial decisions.
 - The correlation patterns across household domains do **not** support a Becker-style division-of-labor story in which women’s low bargaining power in portfolio choice is offset by greater power elsewhere.
 - The same qualitative pattern appears in other countries: the husband’s bargaining weight is about **68.5%** in Germany and about **60.9%** in the United States.
- **Economic takeaway.** Household financial decisions are not well described by either a unitary-household model or a simple “financial head has all the power” rule. Instead, portfolio choice reflects a bargaining process with a sizable and persistent **gender gap**.

2 Data and measurement (key objects)

2.1 Primary data: HILDA and the main sample

- **Main dataset.** The core analysis uses the **Household, Income, and Labour Dynamics in Australia (HILDA)** Survey, a nationally representative Australian panel.
- **Waves used.** The paper builds the main sample from waves **6, 10, 14, and 18**, corresponding to **2006, 2010, 2014, and 2018**.
- **Why these waves.** These waves contain the combination of demographics, household asset allocation, and the identity of the household financial head needed for the analysis. Personality measures are imported from the preceding waves, and cognitive ability is taken from waves 12 and 16 and treated as persistent.
- **Sample restriction.** The baseline sample keeps heterosexual married couples with a husband and a wife, drops households with missing information, drops cases where financial decisions are made by someone outside the household, and drops the rare cases in which both spouses claim to be the sole financial head.
- **Final sample size.** The final HILDA sample contains **9,357 household-wave observations**, representing **4,336 unique households**.

2.2 Observed household- and individual-level variables

- **Portfolio outcomes.**
 - a stock-market participation dummy indicating whether the household directly holds equities,
 - the level of risky-asset holdings for participating households.
- **Household characteristics.** Earnings, total wealth, financial wealth, number of children, and wave indicators.
- **Individual characteristics.** Age, education, employment, earnings, risk aversion, cognitive ability, and Big Five personality traits.
- **Risk-aversion measure.** HILDA asks each individual how much financial risk they are willing to take with spare cash. Responses are coded from **1** to **4**, where a higher number means greater risk aversion.
- **Cognitive ability.** Measured from self-reported math skill and standardized in the data.
- **Financial head.** Each spouse reports who makes decisions about savings, investment, and borrowing. This generates five headship types:
 - husband-headed,
 - husband-shared,
 - jointly headed,
 - wife-shared,
 - wife-headed.

- **Gender norms.** The paper also uses three survey questions on traditional gender norms: division of labor, housework sharing, and the mother’s role.

2.3 Key descriptive facts in the Australian data

- **Household finance facts.**
 - Stock-market participation is about **48%**.
 - Median household income is about **AU\$104,000**.
 - Median total wealth is about **AU\$975,000**.
 - Median financial wealth is about **AU\$241,000**.
 - Mean equity holdings are about **AU\$75,000**, but the median is **zero**.
- **Average within-couple differences.** Husbands are on average about **2.4 years older**, **8 percentage points** more likely to be employed, and earn about **AU\$29,000** more than wives. Wives are on average more risk-averse.
- **Financial-head distribution.** Roughly **58–60%** of households report shared responsibility, about **26–28%** are husband-headed, and about **13–16%** are wife-headed across the HILDA waves used.
- **Heterogeneous spousal risk preferences.** More than **50%** of couples have the same reported risk preference, consistent with assortative matching, but about **43%** of households have spouses with different risk aversion. This gives the paper room to identify bargaining over risk preferences.
- **Simple reduced-form relation.** In linear probability regressions, both spouses’ risk aversion enters negatively and significantly for stock-market participation, confirming that risk preferences matter quantitatively for the portfolio margin under study.

2.4 Why HILDA is especially useful

- The paper argues that HILDA is unusually well suited because it jointly observes:
 - both spouses’ risk preferences,
 - detailed asset allocation,
 - many individual covariates,
 - the identity of the financial head,
 - and later, household-level gender-norm attitudes.
- Other major surveys typically miss at least one of these pieces. That is why the Australian exercise is the paper’s main laboratory, while Germany and the United States are used as supporting evidence.

2.5 External validation datasets: Germany and the United States

- **Germany.** The paper uses the **German Socio-Economic Panel (GSOEP)**. The final sample contains **5,843 household-wave observations** and **3,502 unique households**. The German

data contain only the **extensive margin** of stock-market participation and do not contain the financial-head variable, so the specification is adapted accordingly.

- **United States.** The paper uses the **Health and Retirement Study (HRS)**. The final sample contains **7,014 household-wave observations** and **2,664 unique households**. The HRS covers households age 50 and above, uses hypothetical income gambles to measure risk aversion, and provides less information on the household-decision structure than HILDA.

2.6 Unobservables and timing

- **Persistent household heterogeneity.** The bargaining equation includes a household-specific unobserved component μ , modeled as a finite-mixture random effect.
- **Temporary bargaining shock.** The bargaining equation also includes a transitory shock ϵ_t .
- **Measurement error in risk aversion.** Reported individual risk attitudes are noisy signals of true risk aversion and are allowed to be correlated between spouses.
- **Measurement error in assets.** Observed risky-asset holdings are noisy measures of the model-implied holdings.
- **Timing.** The model is static within a wave: given current household wealth, participation cost, bargaining weights, and individual risk preferences, the household chooses whether to hold risky assets and how much to hold.

3 Models and empirical specifications (all equations + interpretation)

3.1 Portfolio environment

The household can allocate wealth w between a risk-free asset paying return r_f and a risky asset with return $r_f + e_x$, where

$$e_x \sim N(r_x, \sigma_x^2).$$

If the household invests a risky amount a , it pays a one-time participation cost C_i whenever $a > 0$.

Interpretation. The model is a static portfolio-choice problem in the spirit of Merton, augmented with a participation cost so that it can fit both the extensive and intensive margins of household stockholding.

3.2 Household utility and optimal risky holding

The household solves

$$U_i(a) = \max_a \left[w(1 + r_f) + \left(ar_x - C_i - \frac{\gamma_i}{2} a^2 \sigma_x^2 \right) I(a > 0) \right]. \quad (1)$$

The solution is

$$a = \begin{cases} 0, & \gamma_i > \frac{r_x^2}{2\sigma_x^2 C_i}, \\ \frac{r_x}{\gamma_i \sigma_x^2}, & \gamma_i \leq \frac{r_x^2}{2\sigma_x^2 C_i}. \end{cases} \quad (2)$$

- γ_i is household risk aversion.
- C_i is household participation cost.

Interpretation. Households first face a participation cutoff. If risk aversion is too high relative to the participation cost, they stay out of the stock market. If they participate, risky holdings are inversely proportional to risk aversion. This is what allows the extensive and intensive margins to identify different parts of the model.

3.3 Participation cost specification

The participation cost is modeled as a flexible function of household characteristics:

$$\begin{aligned} C_i = & c_0 + c_1 \log(\text{earning})_i + c_2 \log^2(\text{earning})_i + c_3 \log(\text{wealth})_i + c_4 \log^2(\text{wealth})_i \\ & + c_5 \text{age}_i + c_6 \text{age}_i^2 + c_7 \text{education}_i + c_8 \text{cognition}_i + c_9 \text{child}_i \\ & + c_{10} \text{year2010} + c_{11} \text{year2014} + c_{12} \text{year2018}. \end{aligned} \quad (3)$$

Interpretation. The paper treats participation cost as an absorbing reduced-form term that captures any non-risk-preference force shaping stock-market participation: information costs, hassle costs, cognitive costs, family constraints, or other omitted barriers.

3.4 Intrahousehold aggregation of risk aversion

Household risk aversion is aggregated from spouses' individual risk aversions as

$$\frac{1}{\gamma} = \frac{\beta^h}{\gamma^h} + \frac{\beta^w}{\gamma^w}, \quad \beta^h + \beta^w = 1. \quad (4)$$

- γ^h and γ^w are the husband's and wife's true risk aversion.
- β^h and β^w are their bargaining weights.

Interpretation. A spouse with more bargaining power has more ability to make the household portfolio reflect their own risk tolerance. The use of reciprocals of risk aversion is convenient because, under the paper's assumptions, it is equivalent to a collective-bargaining representation in which the household utility function is a weighted average of individual utilities.

3.5 Bargaining power equation

The husband's bargaining power is a logistic transform of a latent index:

$$\beta^h(X_t^h, X_t^w, H_t) = \frac{\exp\left(\tilde{\beta}(X_t^h, X_t^w, H_t)\right)}{\exp\left(\tilde{\beta}(X_t^h, X_t^w, H_t)\right) + 1}, \quad (5)$$

where

$$\tilde{\beta}(X_t^h, X_t^w, H_t) = (X_t^h - X_t^w)\delta_x + \sum_{j=1}^5 \delta_{jt}^H I(H_t = j) + \mu + \epsilon_t. \quad (6)$$

- $X_t^h - X_t^w$ are spousal differences in observed characteristics such as age, education, employment, earnings, cognition, and personality traits.
- H_t is the household type based on the financial-head response.
- δ_x measures the contribution of observed differences.
- δ_{jt}^H captures time-varying gender effects tied to headship type.
- μ is persistent household heterogeneity.
- $\epsilon_t \sim N(0, \sigma_\epsilon^2)$ is a temporary shock.

Interpretation. The bargaining equation separates the part of bargaining power explained by measurable spousal asymmetries from a residual gendered component. This decomposition is central for the paper’s headline result that only about half of the Australian gender gap is explained by observables.

3.6 Measurement equations for risk aversion and assets

Because reported risk attitudes may be noisy, the model maps survey responses into latent true risk aversion:

$$\log \gamma_t^h = \zeta_0^h + \zeta_1^h \log \bar{\gamma}_t^h + \xi_t^h, \quad \log \gamma_t^w = \zeta_0^w + \zeta_1^w \log \bar{\gamma}_t^w + \xi_t^w, \quad (7)$$

with correlated measurement errors

$$\begin{pmatrix} \xi_t^h \\ \xi_t^w \end{pmatrix} \sim N \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \rho_\xi \\ \rho_\xi & 1 \end{pmatrix} \sigma_\xi^2 \right).$$

Observed risky holdings are also noisy:

$$\log \bar{a}_t = \log a_t + \epsilon_t^a, \quad \epsilon_t^a \sim N(0, \sigma_a^2). \quad (8)$$

Interpretation. The model does not take survey responses at face value. Instead, it treats them as imperfect measurements of underlying preference objects.

3.7 Likelihood and panel structure

For a household of unobserved type k , the period- t likelihood is

$$l_t^k = \int_{\xi_t} \int_{\epsilon_t} 1(\tilde{d} = d_t \mid \Omega_t, \epsilon_t, \xi_t, \mu_k) P_a(\tilde{a} = \bar{a}_t \mid \Omega_t, \epsilon_t, \xi_t, \mu_k)^{1(d_t=1)} d\epsilon_t d\xi_t. \quad (9)$$

The full likelihood is

$$L = \prod_i \left[\sum_{k=1}^K p_k \prod_{t \in \{2006, 2010, 2014, 2018\}} l_{it}^k \right]. \quad (10)$$

- The paper uses a finite-mixture distribution for μ with $K = 4$ mass points.
- Estimation proceeds by maximum likelihood, with standard errors from the BHHH algorithm.

Interpretation. The panel dimension helps identify persistent household heterogeneity. The finite-mixture specification lets the model fit rich cross-sectional heterogeneity without estimating thousands of household fixed effects.

3.8 Identification logic

- **Extensive and intensive margins jointly identify risk aversion and participation costs.** Conditional on participating, equation (2) maps risky holdings into household risk aversion. The lowest positive risky holdings pin down the participation threshold and therefore the level of the participation cost.
- **Bargaining coefficients are identified from how inferred household risk aversion varies with spousal characteristics and headship structure.**
- **Measurement equations are identified by comparing latent risk aversion inferred from portfolio choices with reported survey risk attitudes.**

Interpretation. The model's revealed-preference content is doing the identifying work: observed household portfolios reveal latent household risk aversion, and then the model asks which spouse's risk attitude the household behavior is closer to.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in simpler approaches

- **Unitary-household models** ignore disagreement between spouses and therefore cannot measure intrahousehold bargaining power.
- **Survey-based financial-head measures** treat the reported decision-maker as if they had full control, but responsibility does not necessarily equal influence.
- **Reduced-form regressions** can show that certain individual characteristics matter, but they do not translate those effects into structural bargaining weights and make it hard to compare multiple channels quantitatively.

4.2 What identifies the structural approach

- Variation in **risky-asset holdings** among participating households identifies latent household risk aversion.
- Variation in **participation decisions** identifies the participation-cost component.
- Variation in **spousal differences** in earnings, employment, cognition, age, education, and personality shifts the household portfolio through bargaining weights rather than directly through the asset-demand equation.
- Variation in **household head type** identifies the residual gender-effect component in bargaining power.
- The **panel structure** identifies a persistent household component μ separately from transitory shocks.

4.3 Why both the extensive and intensive margins matter

- A model fitted only to participation would confound risk aversion with participation costs.
- The intensive margin gives a direct mapping from holdings to risk aversion once households are inside the market.
- The extensive margin tells the model where the participation cutoff lies. Together, the two margins let the paper disentangle preference heterogeneity from nonparticipation frictions.

4.4 Why the estimates are credible

- The model fits the observed stock-participation and risky-holding moments well across household-head types and risk-preference groupings.
- The paper allows for **correlated measurement errors** in spouses' reported risk aversion and shows that the main gender-gap result survives richer specifications in which spouses' reported risk attitudes partly load on each other's true risk preferences and on marriage duration.
- The paper checks external validity by replicating the main qualitative pattern in **Germany** and the **United States**.

4.5 Main limitations

- The portfolio-choice model is **static**, so it does not analyze dynamic saving, long-run portfolio adjustment, or bargaining over intertemporal preferences.
- The model assumes **full commitment** within marriage and does not explicitly model divorce threats.
- Bargaining is only over **risk preferences**; disagreement over beliefs or return expectations is not modeled.
- The paper identifies welfare implications through the portfolio margin, but it does not directly observe ex post investment performance by bargaining type.

5 Tables and figures

5.1 Figure 1 and Tables 1–4: the data already hint at bargaining heterogeneity

- **Figure 1** shows that most households report shared financial responsibility, but husband-headed households are much more common than wife-headed households.
- **Table 1** shows the baseline asymmetries inside couples: husbands are older, more employed, higher earning, and less risk-averse on average.
- **Table 2** shows that while spouses often sort on risk preferences, disagreement is still common: about **43%** of couples have different risk aversion.
- **Table 3** shows that higher risk aversion is associated with lower stock-market participation for both spouses.

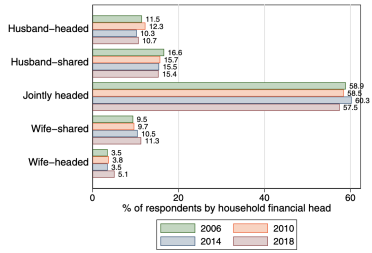


Figure 1
Distribution of the financial head of the household by years
This figure shows the distribution of five household types in four different years (2006, 2010, 2014, 2018). The first types of households are "husband-headed," in which both spouses report the husband makes financial decisions; "jointly headed," in which both husband and wife report that financial decisions are shared equally between the spouses; "wife-headed," in which both spouses report the wife makes financial decisions; "husband-shared," in which one spouse reports "husband" and the other reports "shared equally"; and "wife-shared," in which one spouse reports "wife" and the other reports "shared equally."

Table 1
Summary statistics

	Mean	SD	P25	P50	P75	Husband	Wife	Diff.
<i>Household characteristics</i>								
Stock participation	0.48	0.50	0	0	1			
Household earnings (AUS\$1,000)	119	108	51	104	158			
Total wealth (AUS\$1,000)	1,431	1,602	560	975	1,703			
Financial asset (AUS\$1,000)	504	899	99	241	568			
Equity (AUS\$1,000)	75	303	0	0	20			
Number of children	0.82	1.09	0	0	2			
<i>Individual characteristics</i>								
Age	49.72	15.04	38	49	61	50.92	48.51	2.42***
Education	12.96	2.57	12	12	15	13.01	12.92	0.10*
Employment	0.64	0.48	0	1	1	0.68	0.60	0.08***
Earnings (AUS\$1,000)	47	57	0	36	73	61	33	29***
Risk aversion	3.30	0.67	3	3	4	3.18	3.42	-0.24***
Cognitive ability	0.15	0.85	-0.47	0.29	0.80	0.33	-0.04	0.37***
Extraversion	4.42	1.98	3.67	4.50	5.17	4.50	4.55	-0.05***
Agreeableness	5.42	0.86	5.00	5.50	6.00	5.19	5.66	-0.47***
Conscientiousness	5.27	0.97	4.67	5.33	6.00	5.17	5.36	-0.20***
Stability	5.28	1.03	4.67	5.33	6.00	5.27	5.29	-0.03
Openness	4.19	1.00	3.50	4.17	4.83	4.26	4.13	0.12***

This table reports summary statistics of our main sample. Stock participation is a dummy variable that indicates whether a household directly holds any equities. Both age and education are measured in years. Employment is a dummy variable indicating whether or not an individual is currently employed. Risk aversion is measured using an integer from 1 to 4, with a higher number indicating more risk aversion. Cognitive ability is measured by standardized scores of self-reported math skills. Extraversion, Agreeableness, Conscientiousness, Stability, and Openness are based on 36 personality questions, and their values range from 1 to 7 (see Section A.3 of the Internet Appendix for more details). US\$1 = AUS\$1.2. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 2
Percentage of households by the risk preference of each spouse

		1	2	3	4	Total
Husband	1	0.1	0.4	0.7	0.7	1.9
	2	0.2	1.8	6.4	2.9	11.4
	3	0.3	2.2	30.3	20.3	53.1
	4	0.2	0.5	8.2	24.8	33.6
	Total	0.8	4.9	45.6	48.8	100.0

This table shows the distribution of husband-wife-paired risk aversion. Risk aversion is measured using an integer from 1 to 4, where a higher number indicates a higher level of risk aversion. Each cell reports the fraction of households with a given pair of risk preferences. The off-diagonal terms represent the cases in which the husband and the wife have different risk preferences.

full analysis in any of these data sets. However, it is still feasible to repeat the main part of our estimation using two data sets: HRS for the United States and GSOEP for Germany. We will discuss these exercises later in Section 5.

Table 3
OLS regression of the stock market participation rate on risk preferences

	Couples		Singles	
	(1)	(2)	(3)	(4)
Risk aversion	-0.112*** (0.008)	-0.081*** (0.007)	-0.132*** (0.014)	-0.103*** (0.013)
Risk aversion (wife)	-0.129*** (0.009)	-0.090*** (0.008)		
Age/10		-0.087** (0.042)		-0.029 (0.024)
Age/10, squared		0.010** (0.004)		0.007** (0.003)
Age/10 (wife)		0.023 (0.042)		
Age/10 (wife), squared		0.003 (0.004)		
Education		0.005** (0.002)		0.014*** (0.003)
Education (wife)		0.001 (0.002)		
No. children in HH		-0.003 (0.005)	-0.011 (0.008)	
log HH earning		0.038*** (0.007)	0.029*** (0.009)	
log HH earning, squared		-0.001 (0.001)	-0.002** (0.001)	
log net wealth		-0.067*** (0.023)	-0.114*** (0.013)	
log net wealth, squared		0.006*** (0.001)	0.009*** (0.001)	
2010		-0.066*** (0.015)	-0.029 (0.022)	
2014		-0.155*** (0.014)	-0.060*** (0.020)	
2018		-0.202*** (0.014)	-0.079*** (0.020)	
Female			0.037 (0.084)	-0.123* (0.073)
Female × Risk aversion			-0.022 (0.023)	0.027 (0.020)
Constant	1.280*** (0.031)	0.512*** (0.155)	0.726*** (0.050)	0.555*** (0.090)
Observations	9,357	9,357	3,213	3,213
R ²	.069	.206	.061	.251

This table analyzes the impact of risk aversion on stock market participation. The dependent variable is a dummy equal to one if the household directly holds any equities. *Risk aversion* is measured using an integer from 1 to 4, with a higher number indicating more risk aversion. *No. children in HH* is the number of children in the household. Columns 1 and 2 concern the regressions on married couple households. Columns 3 and 4 repeat the analysis for single households. In these two columns, *Female* is a dummy equal to one if the individual is a female. Robust standard errors are in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

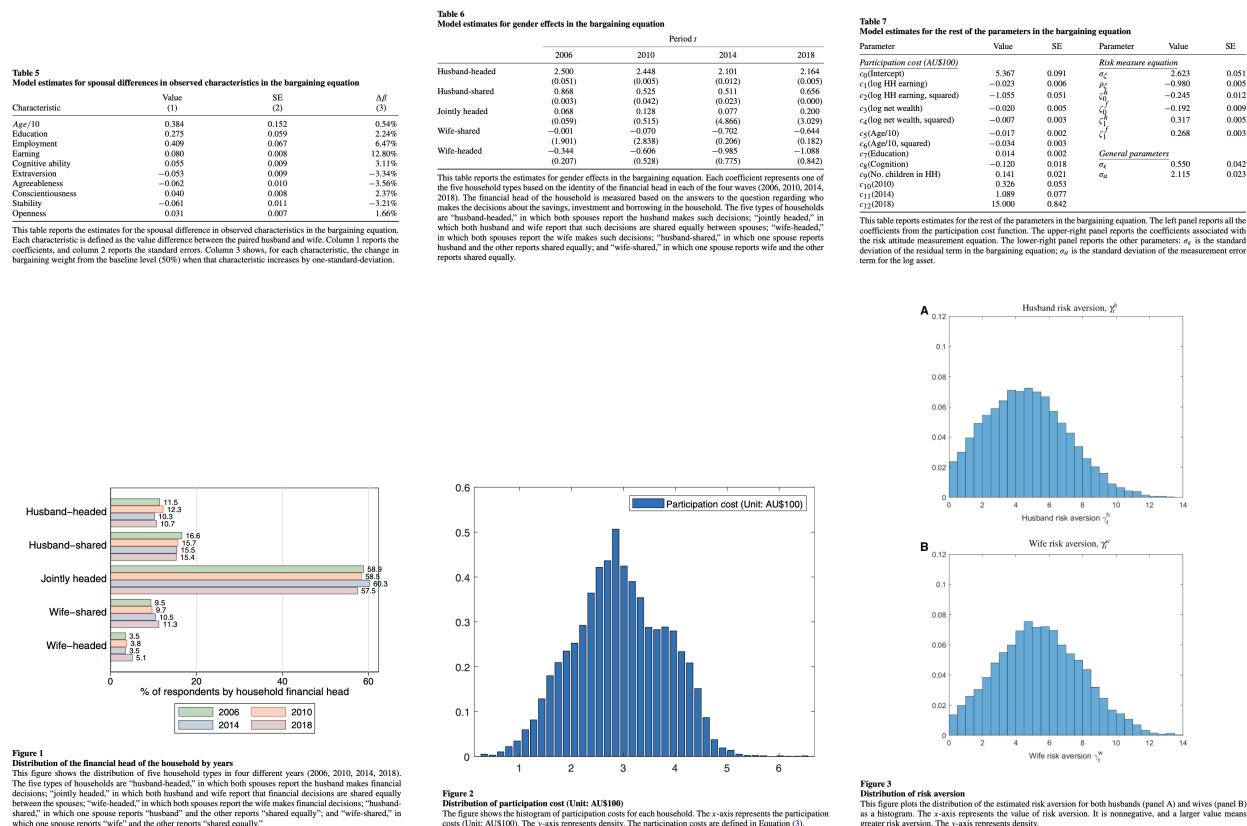
Table 4
Household characteristics by financial head

	Husband-headed				Jointly headed				Wife-headed			
	All	Husband	Wife	Diff.	All	Husband	Wife	Diff.	All	Husband	Wife	Diff.
<i>Individual characteristics</i>												
Age	50.79	52.19	49.39	2.80***	50.21	51.38	49.03	2.35***	49.06	50.34	47.78	2.56*
Education	13.57	13.95	13.20	0.75***	12.80	12.83	12.77	0.06	12.58	12.13	13.03	-0.89***
Employment	0.58	0.66	0.49	0.17***	0.64	0.67	0.60	0.08***	0.61	0.63	0.59	0.04
Earnings (AUS\$1,000)	52	80	25	55***	45	57	32	25***	42	45	39	6
Risk aversion	3.11	2.85	3.36	-0.51***	3.35	3.25	3.45	-0.19***	3.35	3.37	3.33	0.04
Cognitive ability	0.22	0.62	-0.19	0.81***	0.13	0.29	-0.03	0.31***	0.10	0.03	0.16	-0.13*
Extraversion	4.31	4.18	4.45	-0.27***	4.43	4.33	4.53	-0.20***	4.46	4.33	4.59	-0.23**
Agreeableness	5.38	5.06	5.69	-0.62***	5.45	5.23	5.67	-0.44***	5.32	5.10	5.55	-0.45***
Conscientiousness	5.27	5.33	5.22	0.11*	5.29	5.18	5.40	-0.22***	5.07	4.74	5.40	-0.66***
Stability	5.20	5.27	5.13	0.15**	5.33	5.31	5.36	-0.05*	5.16	5.07	5.25	-0.18*
Openness	4.26	4.37	4.15	0.22***	4.16	4.22	4.11	0.11***	4.23	4.21	4.26	-0.05
<i>Household characteristics</i>												
Stock participation	0.64				0.45				0.42			
HH earnings (AUS\$1,000)	144				109				108			
Total wealth (AUS\$1,000)	2,181				1,285				1,309			
Financial asset (AUS\$1,000)	888				440				426			
Equity (AUS\$1,000)	200				47				105			
No. children in HH	0.95				0.78				0.86			

This table reports household characteristics by the three types of financial heads of the household. The financial head of the household is measured based on the answers to the question of who makes the decisions about the saving, investment and borrowing in the household. The three types of households shown here are "husband-headed," in which both spouses report the husband makes financial decisions; "jointly headed," in which both husband and wife report that financial decisions are shared equally between spouses; and "wife-headed," in which both spouses report the wife makes financial decisions. US\$1 = AUS\$1.2. * $p < .1$; ** $p < .05$; *** $p < .01$.

- **Table 4** shows that husband-headed households are richer, participate more in the stock market, and exhibit stronger husband advantages in earnings, employment, and cognition.
- **Read:** These exhibits motivate the model by showing that there is both within-household disagreement and systematic cross-household variation in who appears to dominate financial decisions.

5.2 Tables 5–7 and Figures 2–4: structural estimates and model fit



- **Table 5** shows that employment, earnings, and cognitive ability are the strongest observed determinants of bargaining power. A one-standard-deviation change in earnings raises the husband's bargaining weight by about **12.8%**; employment raises it by about **6.5%**; cognition by about **3.1%**.
- **Table 6** shows large gender effects by headship type. Husband-headed households have strongly positive gender-effect coefficients, whereas wife-headed households have negative ones.
- **Table 7** shows that participation costs decline with earnings, wealth, age, and cognition, and rise with the number of children. The average participation cost is around **AU\$300**.
- **Figure 2** plots a wide distribution of participation costs, indicating substantial heterogeneity beyond risk aversion.
- **Figure 3** shows that estimated male and female risk-aversion distributions differ: husbands have a lower median and more right skew, consistent with the gender-risk literature.

- **Table 8** and **Figure 4** show that the model fits the distribution of stock-market participation and risky holdings reasonably well.
- **Read:** The model is not only statistically estimated; it also reproduces the portfolio moments it is designed to explain.

5.3 Table 9 and Figure 5: decomposition of bargaining power

Table 9
Source of bargaining power heterogeneity

	Bargaining weight (β^H)		Stock participation (3)	Risky asset (log values) (4)
	Mean (1)	SD (2)		
Equal weight ($\beta=0.5$)	0.500	0.000	0.448	9.97
All heterogeneity	0.600	0.226	0.491	10.11
Gender effects (δ^H)	0.567	0.147	0.465	10.03
All observed variables (δ_k)	0.551	0.146	0.464	10.00
Age	0.520	0.036	0.450	9.97
Education	0.513	0.343	0.532	10.36
Employment	0.542	0.255	0.510	10.63
Earning	0.562	0.242	0.497	10.21
Cognitive ability	0.540	0.116	0.461	9.99
Extraversion	0.470	0.179	0.465	10.02
Agreeableness	0.485	0.036	0.448	9.97
Conscientiousness	0.496	0.026	0.447	9.97
Stability	0.500	0.022	0.450	9.97
Openness	0.498	0.026	0.448	9.97

This table quantifies the importance of various components of the model in explaining the distribution of bargaining power between husband and wife. Our method is as follows: we change one particular component each time in the bargaining equation and simulate the bargaining weight and financial decisions (stock market participation and the level of risky asset holdings) for each household. Columns 1 and 2 report the mean and standard deviation of the simulated bargaining weights. Columns 3 and 4 report the average stock market participation and average risky asset holdings. The first line presents the benchmark case, in which spouses have an equal say about financial decisions with the bargaining power of $\beta=0.5$ in our model. The next line presents the full heterogeneity case in which we consider both gender effects and spousal differences in observable characteristics. The next two lines present the cases in which we consider only gender effects and only spousal differences in observable characteristics. The rest of the table reports the importance of each variable one by one.

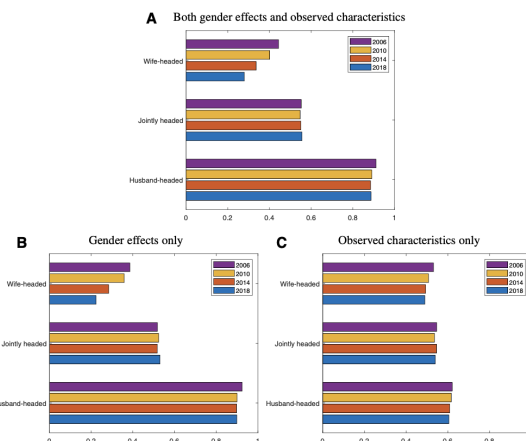


Figure 5
Husband's bargaining power, by financial head structure
The figure plots the average bargaining power a husband has across household head types. "Husband-headed" represents the group in which both spouses report the husband makes financial decisions; "jointly headed" represents the group in which both spouses report that financial decisions are "shared equally" between spouses; and "wife-headed" represents the group in which both spouses report the wife makes financial decisions. Panel A plots, for each household type in any given wave, the average bargaining power a husband has in the baseline model. Panel B reports the simulated average husband bargaining weight when we retain gender effects, and panel C reports the simulated average husband bargaining weight when we retain the heterogeneity from the observed characteristics.

- **Table 9** is the core decomposition table. Under equal weighting, the husband's bargaining power is mechanically **0.50**. In the full model, it rises to about **0.60**.
- When only **gender effects** are active, the husband's average bargaining weight is about **0.567**. When only **observed spousal differences** are active, it is about **0.551**.
- This is the basis for the paper's claim that about **half** of the Australian gap is explained by observables and about half by gender effects.
- **Figure 5** shows that bargaining power rises monotonically from wife-headed to jointly headed to husband-headed households, and that most of the cross-household variation across these groups comes from the **gender-effect** component rather than from observables alone.
- **Read:** The survey-based "financial head" measure is directionally right but quantitatively too extreme. The financial head has more power, but not all the power.

5.4 Tables 10 and 11: robustness to correlated spousal risk preferences

- **Table 10** allows each spouse's reported risk preference to depend partly on the other spouse's true risk preference and, in the most flexible version, on marriage duration.
- The likelihood improves as the specification becomes richer, which supports the idea that spouses' reported risk attitudes are correlated.

Table 10
Parameter estimates under alternative heterogeneity specifications

	(1) <i>Baseline</i>	(2) <i>Intermediate</i>	(3) <i>Full</i>
ζ_0^h	-0.236 (0.013)	-0.429 (0.017)	-0.348 (0.013)
ζ_0^w	-0.195 (0.011)	-0.527 (0.016)	-0.515 (0.019)
ζ_1^h	0.317 (0.005)	0.304 (0.005)	0.229 (0.004)
ζ_1^w	0.267 (0.003)	0.281 (0.004)	0.296 (0.004)
ζ_2^h		0.042 (0.003)	0.031 (0.002)
ζ_2^w		0.035 (0.002)	0.033 (0.003)
ζ_{1d}^h			-0.0099 (0.0004)
ζ_{1d}^w			-0.0045 (0.0004)
ζ_{2d}^h			0.0018 (0.0002)
ζ_{2d}^w			0.0030 (0.0003)
σ_ϵ	2.627 (0.061)	2.394 (0.059)	2.671 (0.064)
ρ_ϵ	-0.980 (0.008)	-0.980 (0.007)	-0.980 (0.005)
Obs.	8,508	8,508	8,508
log L	-14,600	-14,531	-14,444
LR tests		(1) & (2)	(2) & (3)
p -value		<.001	<.001

This table reports estimates of three nested model specifications. The *Baseline* case is when $\zeta_{1d}^h = \zeta_{1d}^w = \zeta_2^h = \zeta_2^w = \zeta_{2d}^h = \zeta_{2d}^w = 0$, which reduces to the original specification in Section 3.2. In the *Intermediate* case, represented by $\zeta_{1d}^h = \zeta_{2d}^h = \zeta_{1d}^w = \zeta_{2d}^w = 0$, an individual's reported risk preference is influenced not only by their true risk preference but also by their spouse's true risk preference. However, the coefficients for the true risk preferences remain constant in this case. Lastly, the *Full* case maintains the *Intermediate* model's structure but with coefficients for the true risk preferences varying based on the duration of the marriage. Standard errors are reported in parentheses. The row labeled log L displays the likelihood values for the three specifications. The “ p -value” row presents the results of the likelihood ratio (LR) test, evaluating the current specification against the preceding one.

- **Table 11** shows that even under these richer specifications, the husband's average bargaining weight remains well above one half: around **0.598** in the baseline and intermediate specifications and about **0.621** in the full specification.
- **Read:** The gender gap is not an artifact of naively taking self-reported risk attitudes literally.

5.5 Tables 12 and 13: gender norms and the division-of-labor hypothesis

Table 12
Gender norms and bargaining weights

	Bargaining weight of husbands		Husband-headed
	(1) Only gender norms	(2) With controls	(3) With controls
Division of labor (husband)	0.001 (0.001)	0.003*** (0.001)	0.004** (0.002)
Share housework (husband)	0.004*** (0.001)	-0.000 (0.001)	0.009*** (0.002)
Mother's role (husband)	-0.001 (0.001)	0.001 (0.001)	0.011*** (0.003)
Division of labor (wife)	0.005*** (0.001)	0.004*** (0.001)	-0.001 (0.003)
Share housework (wife)	-0.002 (0.001)	-0.001 (0.001)	0.003 (0.002)
Mother's role (wife)	0.005*** (0.001)	0.002** (0.001)	0.007*** (0.002)
Observations	8,378	8,378	8,378

This table analyzes the impact of gender norms on bargaining weights. Attitudes about gender norms are measured by three questions that elicit participants' attitudes toward the division of labor, the share of housework in the family, and the mother's role. Answers to each question are measured on a 7-point Likert scale from 1 (strongly disagree) to 7 (strongly agree). We recode all variables so that a higher value represents a more traditional view of gender norms (see [Section A.5](#) of the [Internet Appendix](#) for more details). The dependent variable in columns 1 and 2 is the simulated bargaining weight from in the bargaining equation, which is the same as the one shown in Figure 5, panel A. Column 1 reports the results of a simple OLS regression of this simulated bargaining weight of husbands on both husbands' and wives' gender norm questions. Column 2 includes extra controls: age, education, household income, household wealth, cognitive ability, and personality traits. Column 3 regresses the dummy variable indicating a husband-headed household on perceived gender norms. Robust standard errors are in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

- **Table 12** links more traditional gender norms to more bargaining power for husbands and a greater probability that the household is husband-headed in financial decisions.
- Both the husband's and the wife's stated norms matter, which the paper interprets as evidence that changing norms on both sides of the household is relevant.

Table 11
The distributions of bargaining weights

	(1) <i>Baseline</i>	(2) <i>Intermediate</i>	(3) <i>Full</i>
10%	0.287	0.180	0.167
25%	0.431	0.328	0.383
50%	0.604	0.642	0.688
75%	0.781	0.872	0.882
90%	0.909	0.951	0.961
Mean	0.598	0.598	0.621

The distribution of simulated bargaining weights attributed to husbands across the three specified specifications.

Table 13
Correlation between responses to household investment decisions and other household decisions

Domain	Correlation with savings, investment, and borrowing
Managing day-to-day spending and paying bills	0.52
Making large household purchases	0.53
The number of hours spent in paid work	0.22
The number of hours partner/spouse spent in paid work	0.12
The way children are raised	0.04
Social life and leisure activities	0.04

This table investigates the correlation between household investment decisions and other household decisions. In addition to asking about household decision-making in “Savings, investment and borrowing,” the HILDA Survey also asks about household decision-making along six other domains: (1) managing day-to-day spending and paying bills; (2) making large household purchases (e.g., cars and major appliances); (3) the number of hours spent in paid work; (4) the number of hours partner/spouse spent in paid work; (5) the way children are raised; and (6) social life and leisure activities. Respondents are given the following options: themselves, their spouses, shared equally between spouses, or other people. In each domain, we classify all households into five types: “husband-headed,” in which both spouses report the husband makes such decisions; “jointly headed,” in which both husband and wife report that such decisions are shared equally between the spouses; “wife-headed,” in which both spouses report the wife makes such decisions; “husband-shared,” in which one spouse reports husband and the other reports shared equally; and “wife-shared,” in which one spouse reports wife and the other reports shared equally.

Table 14
Comparison of bargaining power heterogeneity between Australia, Germany, and the United States

	Australia		Germany		US	
	Bargaining weight (β^H) (1)	Stock participation (2)	Bargaining weight (β^H) (3)	Stock participation (4)	Bargaining weight (β^H) (5)	Stock participation (6)
Equal weight ($\beta=0.5$)	0.500 [0.500,0.500]	0.429 [0.399,0.496]	0.500 [0.500,0.500]	0.266 [0.180,0.231]	0.500 [0.500,0.500]	0.352 [0.204,0.496]
All heterogeneity	0.600 [0.580,0.611]	0.491 [0.444,0.539]	0.685 [0.582,0.788]	0.279 [0.240,0.309]	0.609 [0.507,0.711]	0.384 [0.252,0.507]
Gender effects (β^G)	0.567 [0.558,0.577]	0.465 [0.416,0.514]	0.837 [0.765,0.909]	0.232 [0.204,0.260]	0.622 [0.541,0.703]	0.356 [0.207,0.498]
All observed variables (δ_x)	0.551 [0.544,0.557]	0.464 [0.417,0.510]	0.549 [0.504,0.594]	0.238 [0.213,0.263]	0.552 [0.559,0.607]	0.358 [0.215,0.495]

This table quantifies the distribution of bargaining power between husband and wife in Australia, Germany, and the United States. Our method is as follows: we change one particular component each time in the bargaining equation and simulate the bargaining weight and financial decisions (stock market participation and the level of risky asset holdings) for each household. To obtain the 95% confidence interval (displayed within brackets), we implement a bootstrap method that involves drawing sets of parameters randomly from their estimated distributions. For each set of drawn parameters, we simulate the outcomes, namely, stock market participation, the level of risky asset holdings, and bargaining power, for each household. This process of drawing and simulation is repeated $R=200$ times, providing us with a distribution of sample averages for the three aforementioned outcomes. Subsequently, we calculate the 2.5nd and 97.5th percentiles of this bootstrap distribution, which enables us to construct a confidence interval for each outcome. Columns 1 and 2 report the average bargaining weights and stock market participation rate in Australia. Columns 3 and 4 report the average bargaining weights and market participation rate in Germany. Columns 5 and 6 report the average bargaining weights and market participation rate in the United States. The first two lines present the benchmark case, in which spouses have an equal say about financial decisions with the bargaining power of $\beta=0.5$ in our model. The next two lines present the full heterogeneity case in which we consider both gender effects and spousal differences in observable characteristics in our baseline fully hedged model. The next four lines present the cases in which we consider only gender effects and only spousal differences in observable characteristics.

- **Table 13** shows that financial decision authority is strongly correlated with authority over **day-to-day spending and bills** and **large household purchases**, but much less correlated with labor-supply, child-rearing, or leisure decisions.
- **Read:** This weighs against the idea that women’s low bargaining power in portfolio choice is simply offset by higher bargaining power in consumption or other domains.

5.6 Table 14: evidence from Germany and the United States

- **Germany.** The husband’s bargaining weight is about **0.685**. The gender-effect component is particularly large, around **0.837** when isolated.
- **United States.** The husband’s bargaining weight is about **0.609**. The isolated gender-effect component is about **0.622**.
- **Australia.** For comparison, the corresponding numbers are about **0.600** for total heterogeneity and **0.567** for gender effects alone.
- **Read:** The qualitative result is broad, not country-specific: household financial decisions place more weight on men’s preferences in all three datasets.

6 Short summary

- **Method.** Build a structural household portfolio-choice model in which the household’s risk aversion is a bargaining-weighted aggregation of spouses’ risk aversion, and estimate it by maximum likelihood using both participation and risky-holding data.
- **Identification.** Portfolio choices reveal household risk aversion; spousal risk measures and headship type reveal how that household risk aversion is aggregated from individual risk preferences.
- **Application.** Australian HILDA data, with additional evidence from German GSOEP and U.S. HRS data.
- **Main result.** In the average household, the husband’s preference gets more weight than the wife’s. In Australia the split is about **60–40**; around half of the gap is explained by observable spousal differences and half by gender effects. Similar or larger gaps appear in Germany and the United States.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper shows that bargaining power inside the household can be identified from **revealed portfolio choices**, not only from survey statements about who makes the decisions.
- It establishes a substantial **gender gap in household financial bargaining power**. On average, husbands' risk preferences receive more weight than wives' risk preferences in actual portfolio decisions.
- This gender gap is driven both by **observable asymmetries**—especially earnings and employment—and by a residual **gender effect** linked to financial headship and gender norms.

7.2 Economic intuition

- If household members disagree about risk, the portfolio chosen by the household must reflect a weighted compromise.
- The spouse with more bargaining power can push the household portfolio closer to their own preferred level of risk-taking.
- Because husbands are on average less risk-averse and also receive greater bargaining weight, households invest more in risky assets than they would under an equal-weight benchmark.
- Traditional gender norms can reinforce this pattern by giving husbands extra influence beyond what would be implied by earnings, employment, and other measurable characteristics alone.

7.3 Takeaway

This paper's main lesson is that household finance is not only about preferences and constraints; it is also about **intra-household power**. Portfolio allocations reveal that this power is unequally distributed by gender, and that unequal distribution is quantitatively important for understanding stock-market participation and risky-asset holdings.